

EX.NO : 11

IMPLEMENTING ERROR DETECTING CODE USING CRC – CCITT

AIM :

To write a Java program for Error Detecting code using CRC – CCITT (16 bits).

ALGORITHM :

- 1) Start the program.
- 2) Read the message from the user.
- 3) Calculate and append the CRC remainder to the message.
- 4) Prompt the user to enter the received data and check its CRC remainder.
- 5) Print the result.
- 6) Stop the program.

PROGRAM :

```
import java.util.*;

public class Crc
{
    public static int n;

    public static void main(String[] args)
    {
        Scanner in=new Scanner(System.in);

        Crc ob=new Crc();

        String code, copy, rec,zero="0000000000000000";
```

```

        System.out.println("Enter message");
        code=in.nextLine();
        n=code.length();
        copy=code;
        code+=zero;
        code=ob.divide(code);
        System.out.println("Message="+copy);
        copy=copy.substring(0,n)+code.substring(n);
        System.out.println("CRC=");
        System.out.println(code.substring(n));
        System.out.println("transmitted frame is "+copy);
        System.out.println("Enter received data");
        rec=in.nextLine();
        if(zero.equals(ob.divide(rec).substring(n)))
            System.out.println("Correct bits received");
        else
            System.out.println("Received frame contains one or more
errors");

        in.close();
    }
    public String divide(String s)
    {
        int i,j;
        char x;

```

```
String div="100010000000100001";
for(i=0;i<n;i++)
{
    x=s.charAt(i);
    for(j=0;j<17;j++)
    {
        if(x=='1')
        {
            if(s.charAt(i+j)!=div.charAt(j))

s=s.substring(0,i+j)+"1"+s.substring(i+j+1);

            else

s=s.substring(0,i+j)+"0"+s.substring(i+j+1);

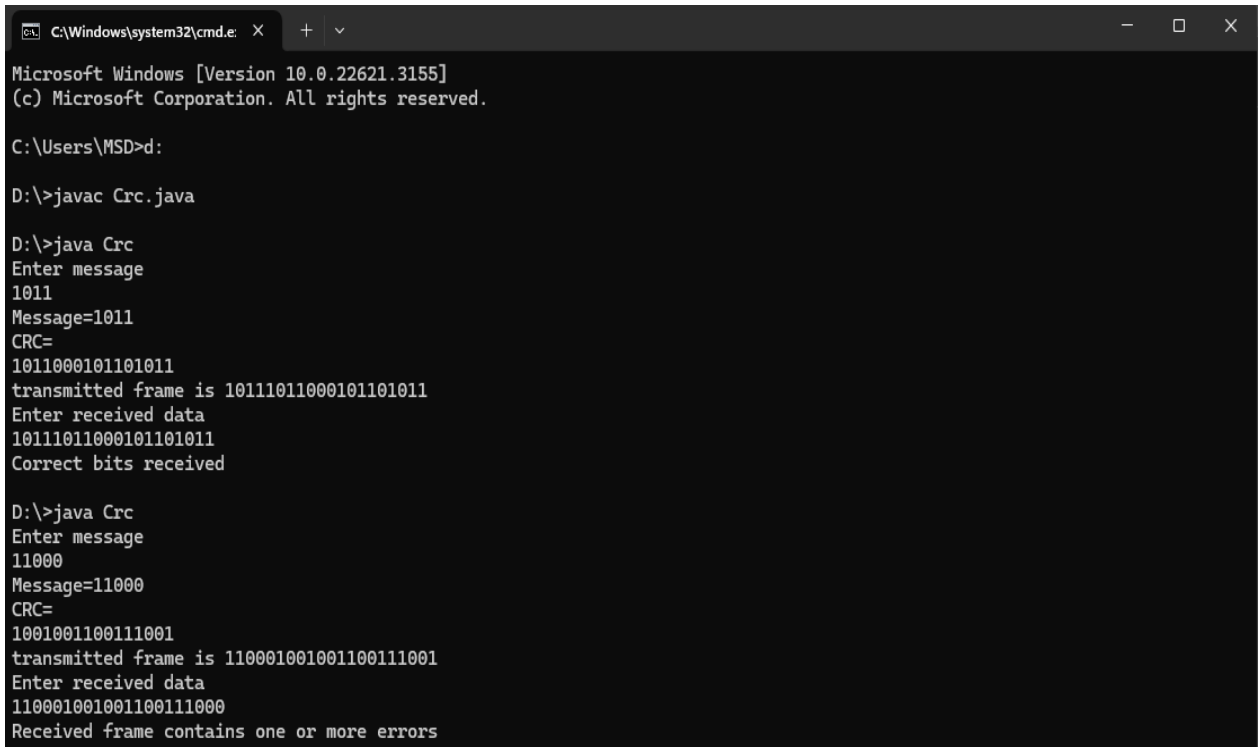
        }

    }

    return s;

}
```

OUTPUT :



```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.22621.3155]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MSD>d:

D:\>javac Crc.java

D:\>java Crc
Enter message
1011
Message=1011
CRC=
1011000101101011
transmitted frame is 10111011000101101011
Enter received data
10111011000101101011
Correct bits received

D:\>java Crc
Enter message
11000
Message=11000
CRC=
1001001100111001
transmitted frame is 110001001001100111001
Enter received data
110001001001100111000
Received frame contains one or more errors
```

RESULT :

Thus, the program for Error Detecting code using CRC – CCITT (16 Bits) is successfully executed and the output is verified.